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Bài 1:
1. Câu lệnh if:

```
if rm $1
then
echo "$1 is deleted sucessfully!"
fi
```

```
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 rmif
trungtin@trungtin-Thin-GF63-12VE:~$ ./rmif aaaa
rm: cannot remove 'aaaa': Is a directory
trungtin@trungtin-Thin-GF63-12VE:~$ ./rmif a
a is deleted sucessfully!
trungtin@trungtin-Thin-GF63-12VE:~$
```

2. Lệnh test và [expr]

```
if test $1 -gt 0
then
echo "$1 is positive number"
fi
```

```
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 ispositive
trungtin@trungtin-Thin-GF63-12VE:~$ ./ispositive 3
3 is positive number
trungtin@trungtin-Thin-GF63-12VE:~$ ./ispositive -3
trungtin@trungtin-Thin-GF63-12VE:~$
```

3. Lệnh if … else … fi

```
if test $1 -gt 0
then
echo "$1 is positive"
else
echo "$1 is negative"
fi
```

```
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 checkNumber
trungtin@trungtin-Thin-GF63-12VE:~$ ./checkNumber 2
2 is positive
trungtin@trungtin-Thin-GF63-12VE:~$ ./checkNumber -5
-5 is negative
trungtin@trungtin-Thin-GF63-12VE:~$
```

5. Lệnh lặp for

VD1:

```
for i in 1 2 3 4 5
do
echo "Welcome $i times"
done
```

```
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 testfor
trungtin@trungtin-Thin-GF63-12VE:~$ ./testfor
Welcome 1 times
Welcome 2 times
Welcome 3 times
Welcome 4 times
Welcome 5 times
trungtin@trungtin-Thin-GF63-12VE:~$
```

VD2:

```
for (( i = 0 ; i <= 5; i++
 ))
do
echo "Welcome $i times"
done
```

```
trungtin@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 for2
trungtin@trungtin-Thin-GF63-12VE:~$ ./for2
Welcome 0 times
Welcome 1 times
Welcome 2 times
Welcome 3 times
Welcome 4 times
Welcome 5 times
trungtin@trungtin-Thin-GF63-12VE:~$
```

VD3:

```
Open  • nestedfor
~/
for (( i = 1; i <= 5; i++ ))    ### Vòng lặp ngoài ###
do
        for (( j = 1 ; j <= 5; j++ )) ### Vòng lặp trong ###
        do
                echo -n "$i "
        done
echo "" #### in dòng mới ###
done
```

```
trungtin@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 nestedfor
trungtin@trungtin-Thin-GF63-12VE:~$ ./nestedfor
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
trungtin@trungtin-Thin-GF63-12VE:~$
```

6. Vòng lặp while:

```
Open  whileViDu
~/
i=1
while [ $i -le 5 ]
do
echo “Welcome $i times”
i=`expr $i + 1`
done
```

```
trungtin@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 whileViDu
trungtin@trungtin-Thin-GF63-12VE:~$ ./whileViDu
“Welcome 1 times”
“Welcome 2 times”
“Welcome 3 times”
“Welcome 4 times”
“Welcome 5 times”
trungtin@trungtin-Thin-GF63-12VE:~$
```

7. Lệnh Case

```
Open    casetest
~/
Please choose a number:
echo "1: Fried Eggs"
echo "2: Salad"
echo "3: Noodle"
read f
case $f in
 "1") echo "Thanks, you chose Fried Eggs.";;
 "2") echo "Thanks, you chose Salad.";;
 "3") echo "Thanks, you chose Noodle";;
 *) echo "Sorry, we have only Fried Eggs, Salad, Noodle";;
esac
```

```
trungtin@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE:~$ chmod 755 casetest
trungtin@trungtin-Thin-GF63-12VE:~$ ./casetest
./casetest: line 1: Please: command not found
1: Fried Eggs
2: Salad
3: Noodle
5
Sorry, we have only Fried Eggs, Salad, Noodle
trungtin@trungtin-Thin-GF63-12VE:~$
```

Bài 2:

Open baitap2_1 ~/

baitap2_1 baitap2_2

```
read -p "Nhap n: " n
for (( i = 1; i <= $n; i++ ))
do
                for (( j = 1 ; j <= i ; j++ ))
                do
                        echo -n "*"
                done
        echo ""
done
```

trungtin@trungtin-Thin-GF63-12VE: ~

```
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap2_1
Nhap n: 5
*
**
***
****
*****
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap2_1
Nhap n: 10
*
**
***
****
*****
******
*******
********
*********
**********
trungtin@trungtin-Thin-GF63-12VE:~$
```

Open baitap2_2 ~/

```
read -p "Nhap n: " n
for (( i = 1; i <= $n; i++ ))
do
        for ((j = 1; j <= $n; j++))
        do
                echo -n "$j    "
        done
        echo ""
done
```

```
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap2_2
Nhap n: 5
1    2    3    4    5
1    2    3    4    5
1    2    3    4    5
1    2    3    4    5
1    2    3    4    5
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap2_2
Nhap n: 10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
1    2    3    4    5    6    7    8    9    10
trungtin@trungtin-Thin-GF63-12VE:~$
```

Bài 3:

Open baitap3 ~/

```
read -p "Nhap he so a = " a
read -p "Nhap he so b = " b
if [ "$a" -eq 0 ]
then
        if [ "$b" -eq 0 ]
        then
                echo "Phuong trinh vo so nghiem!!"
        else
                echo "Phuong trinh vo nghiem!!"
        fi
else
        echo -n "Ket qua cua phuong trinh bac 1 la: x = "
        echo "scale=2; (-1) * $b/$a" | bc
fi
```

```
trungtin@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap3
Nhap he so a = 3
Nhap he so b = 7
Ket qua cua phuong trinh bac 1 la: x = -2.33
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap3
Nhap he so a = 0
Nhap he so b = 0
Phuong trinh vo so nghiem!!
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap3
Nhap he so a = 2
Nhap he so b = -9
Ket qua cua phuong trinh bac 1 la: x = 4.50
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap3
Nhap he so a = 2
Nhap he so b = 4
Ket qua cua phuong trinh bac 1 la: x = -2.00
trungtin@trungtin-Thin-GF63-12VE:~$
```

Bài 4:

```
Open    baitap4
        ~/
echo -n "Nhap n: "
read n
S=0;
for (( i = 1; i <= $n ; i++ ))
do
        S=`expr $S + $i`
done
echo "Tong day so nguyen tu 1 den $n la: "
echo "$S"
```

```
trungtin@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap4
Nhap n: 5
Tong day so nguyen tu 1 den 5 la:
15
trungtin@trungtin-Thin-GF63-12VE:~$
```

Bài 5:

Open baitap5 ~/

```
while true
do
        echo -n "Nhap so nguyen n: "
        read n
        for (( i = 1 ; i <= 10; i++ ))
        do
                echo "$i x $n = `expr $i \* $n`"
        done
        echo -n "Ban co muon tiep tuc khong? (y/n)?"
        read cont
        if [ "$cont" != "y" ]
        then
                echo "Ket thuc chuong trinh"
                break
        fi
done
```

```
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap5
Nhap so nguyen n: 5
1 x 5 = 5
2 x 5 = 10
3 x 5 = 15
4 x 5 = 20
5 x 5 = 25
6 x 5 = 30
7 x 5 = 35
8 x 5 = 40
9 x 5 = 45
10 x 5 = 50
Ban co muon tiep tuc khong? (y/n)?y
Nhap so nguyen n: 6
1 x 6 = 6
2 x 6 = 12
3 x 6 = 18
4 x 6 = 24
5 x 6 = 30
6 x 6 = 36
7 x 6 = 42
8 x 6 = 48
9 x 6 = 54
10 x 6 = 60
Ban co muon tiep tuc khong? (y/n)?n
Ket thuc chuong trinh
trungtin@trungtin-Thin-GF63-12VE:~$
```

Bài 6:

Open baitap6 ~/

```
read -p "Nhap so thu muc can tao: " n
for (( i = 1; i <= $n ; i++))
do
        echo "Tao thu muc user_$i"
        mkdir user_$i
        if [ "$?" -eq 0 ]
        then
                echo "Da tao thu muc user_$i thanh cong"
        else
                echo "Khong tao duoc thu muc user_$i"
        fi
done
```

trungtin@trungtin-Thin-GF63-12VE: ~

```
trungtin@trungtin-Thin-GF63-12VE:~$ ./baitap6
Nhap so thu muc can tao: 5
Tao thu muc user_1
Da tao thu muc user_1 thanh cong
Tao thu muc user_2
Da tao thu muc user_2 thanh cong
Tao thu muc user_3
Da tao thu muc user_3 thanh cong
Tao thu muc user_4
Da tao thu muc user_4 thanh cong
Tao thu muc user_5
Da tao thu muc user_5 thanh cong
trungtin@trungtin-Thin-GF63-12VE:~$ ls
aaaa        checkNumber          hs_err_pid3574.log    Pictures    user_3
arg1        Desktop              hs_err_pid5408.log    Public      user_4
baitap2_1   Documents            hs_err_pid5889.log    rmif        user_5
baitap2_2   Downloads            ispositive            snap        Videos
baitap3     eclipse-workspace    Learning_Programming  stream.yuv  whileViDu
baitap4     for2                 lmstudio              Templates
baitap5     hocTap               map.txt               testfor
baitap6     hocTap1              Music                 user_1
casetest    hs_err_pid18207.log  nestedfor             user_2
trungtin@trungtin-Thin-GF63-12VE:~$
```